FAQ on library concepts

Q:what IS LIBRARY?

Q:Role of linker , what is Linkage ?
Q:Types of libraries and Common file extensions for static and dynamic libraries in C?

Q:Types of LINKING?

Q:Static linking and Dynamic linking?

Q:What are the common file extensions for static and dynamic libraries in C?

Q:Difference between static libraries and dynamic libraries.

Q:Write commands to creat static library file and dynamic library file?

Q:Process of linking a static library to a C program.

Q: Process of linking a dynamic library (shared library) to a C program.

Q:Role of the linker in linking dynamic libraries?

Q:Advantages and disadvantages of using static libraries.
Q:How does dynamic linking affect the size and performance of C programs?

Q:.A function is statically linked to main, if ----------is known during ----------.

Q:.A function is ----------linked to main ,if refrence of called function is know during runtime
or load time.

Q:Functions from dynamic library file can be statically linked or not?

Q:what is dynamic load time linking and run time linking

Q:Explain dynamic run time linking with program.

Q:.what is call back function

Q:function pointer ,Is array of function pointer possibel or not?

Q:.Printf can accept variable number of arguments ,how printf function has been implemented to accept number of arguments.

Q:.In scanf argument address is required , but in printf we do not require address of argument .Why?

Q:Which of the following is an advantage of using dynamic libraries over static libraries?

a) Smaller executable size

b) Faster program startup time

c) Easier portability across different platforms

d) Easier debugging process

Answer: a) Smaller executable size

Q:Which of the following file extensions is commonly associated with static libraries in C?

a) .so

b) .dll

c) .lib

d) .a

Answer: c) .lib

Q:What is static linking in the context of libraries?

a) Linking library functions at compile time

b) Linking library functions at runtime

c) Dynamically loading libraries into memory

d) Generating library documentation

Answer: a) Linking library functions at compile time

q:When statically linking a library in C, which file is typically included in the compilation process?
a) Source code file (.c)
b) Object file (.o)
c) Dynamic library file (.dll)
d) Static library file (.lib or .a)

Answer: d) Static library file (.lib or .a)

Q:Which command is commonly used to statically link a library during the compilation process in C?

a) gcc -dynamic

b) gcc -shared

c) gcc -static

d) gcc -link

Answer: c) gcc -static

Which of the following is a drawback of static linking?

a) Increased runtime dependencies

b) Higher memory usage

c) Limited portability across different platforms

d) Incompatibility with certain development environments

Answer: b) Higher memory usage

Which of the following statements about dynamic linking is true?

a) It results in a larger executable size compared to static linking.

b) It requires the presence of static libraries at runtime.

c) It allows for the modification of library functions without recompilation. d) It only links library functions at compile time.

Answer: c) It allows for the modification of library functions without recompilation.

Q:When dynamically linking a library in C, which file is typically included during the compilation process?
a) Source code file (.c)

b) Object file (.o)

c) Dynamic library file (.dll or .so)

d) Static library file (.lib or .a)

Answer: c) Dynamic library file (.dll) or .so)

Q:Which command is commonly used to dynamically link a library during the compilation process in C?

a) gcc -dynamic

b) gcc -shared

c) gcc -static

d) gcc -dynamic-link

Answer: b) gcc -shared

Which of the following is an advantage of dynamic linking over static linking?

a) Reduced runtime dependencies

b) Faster program startup time

c) Better control over library updates

d) Smaller executable size

Answer: d) Smaller executable size

Q:

FUNCTION POINTERS

1.What is a function pointer in C?

2. How do you declare a function pointer?

3. How do you assign a function to a function pointer?

4.How do you call a function through a function pointer?

5.How are function pointers useful in callback functions?

6.What happens if a function pointer is dereferenced before being assigned to a function?

7.How are function pointers different from regular pointers?

Syntax of function pointer?

a.int (*fp[])()

b. int *f[](int , int)

c. int (*fp)(const char *,...)

d. int (*fp)[2](int, char)

Q:what is callback function ?

a.returning function address

b.calling function with function pointer is called callback function

c.passing function address as an argument to another function

d.recurssion is also called callback function

Q:Array of function pointer is possible

Select one:

True

False

Q:Which of the statement is correct to have array of function pointer

a.int (*fp[10])(int,int)

b.void *p[10](int,int)

c.Not possible

d.void (*p)[10](int , int)

Variable argument list

1:What is a variable argument list in C?

2:What is the difference between a fixed argument list and a variable argument list?

3:Which header file is required to use variable argument lists in C?

4:What is the role of the <stdarg.h> header file in C programming?Can you use variable argument lists with user-defined functions?

5::How do you declare a function with a variable argument list in C?

6:How do you handle variable types within a function with a variable argument list?

7.What happens if you pass an argument of the wrong type to a function with a variable argument list?

Which header file contains the macros used for variable argument lists in C? a) <stdio.h>

b) <stdlib.h>

c) <stdarg.h>

d) <math.h>

Answer: c) <stdarg.h>

Q:What does the va_arg macro do in C?

a) Initializes a va_list variable

b) Retrieves the next argument from a variable argument list

c) Signals the end of a variable argument list

d) Resets a va_list variable

Q:**Which macro is used to initialize a va_list variable with the variable arguments?**

a) va_start b) va_arg c) va_end d) va_copy

Q:**Which macro is used to retrieve the type of the next argument from a variable argument list in C?**

a) va_start

b) va_arg

c) va_end

d) va_copy

Q:**Which macro is used to initialize a va_list variable in C?**

a) va_start

b) va_arg

c) va_end

d) va_list

PRINTF behaviour

syntax: int printf(const char *format, ...);

Q:What does the printf function return in C?

a) The number of characters to dispaly

b) The value of the last argument passed

c) A pointer to the formatted string

d) The return value of the format string

Answer: a) The number of characters to dispaly

1.int main()

```
{
    int num = 42;
    int res;

    res= printf("hello");
```

```
    printf("Number of characters to dispaly is : %d\n",
res);


    return 0;
}
```

2:

```
int main()
{
    int num = 2;
    int res;
    res=printf("%d",num);
    printf("\n");
    printf("return value is %d",res);
}
```

Q:

```
int main()
 {
    int num1 = 2,num2=10;
```

```
    int res;
    res=printf("%d%d",num1,num2);
    printf("\n");
    printf("return value is %d",res);
}
```

Q:

```
int main()
{
    int num1 = 2,num2=10;
    int res;
    res=printf("%d\n%d",num1,num2);

    printf("return value is %d",res);
}
```

Q:

```
  int main()
 {
  printf("%3d\n",2);
  printf("%3d\n",45);
  printf("%4d",45);
 }
```

Q:

```
int main()
{
int num1 = 2, num2 = 10;
int res;
res = printf("%*c%*c", num1, 32, num2, 32);
printf("return value is %d", res);//12
}
```

SCAN F BEHAVIOUR

syntax:int scanf(const char *format, ...);

Q.What does the scanf function return in C?

a) The number of characters read

b) The value of the last input item read

c) A pointer to the input buffer

d) The number of successful assignments made

Answer: d) The number of successful assignments made

Q:int main()

{

int num1, num2;

int res;

printf("Enter two integers\n ");

res = scanf("%d%d", &num1, &num2);

printf("Return value is %d\n", res);//2

}

Q:int main()

{

int num1, num2,num3;

int res;

printf("Enter two integers\n ");

res = scanf("%d%d%d", &num1, &num2,&num3);

printf("Return value is %d\n", res);

}